



IILM
UNIVERSITY

Bachelor of Computer Science and Engineering

Practical File

Course Name: Software Engineering Lab

Course Code: CSE24

Semester: 4th

Submitted to:
Dr. Tanu Gupta
Assistant Professor

Submitted by:
Shivansee
2481187

School of Computer Science & Engineering
IILM University, Gurugram

INDEX

[illegible]

PRACTICAL – 1

Aim: To prepare a Software Requirements Specification (SRS) document for a Library Management System according to IEEE standards.

Introduction: A Software Requirements Specification (SRS) is a document that explains the requirements of a software system. It describes what the system should do and how it should work. The Library Management System (LMS) is developed to manage library activities such as adding books, issuing books, returning books, managing users, and keeping records in a digital format. This system helps librarians handle library operations easily and efficiently. This document is useful for developers, testers, project managers, and users who are involved in developing and using the system.

Case Tool: StarUML

Scenario 1: Library Management System

Actors Involved:

1. Administrator
2. Student
3. Faculty

Use Case for Each one:

1. Administrator

1.1 Register

The administrator creates an account in the system to access administrative features.

1.2 Login

The administrator logs into the system using username and password to manage the system.

1.3 Update Book Status

The admin can change the status of books such as issued, returned, lost, or damaged.

1.4 Issue Book

The admin checks requests from users and issues books to students or faculty members.

1.5 Search

The admin can search the database to find books or user information.

1.6 Calculate Fine

The admin calculates a fine if a user returns a book after the due date.

1.7 Add or Remove User

The admin can add new users to the system or remove existing users.

2. Student

2.1 Register

The student creates an account in the system to access library services.

2.2 Login

The student logs into the system to view and manage library activities.

2.3 Search Books

The student searches for books using title, author name, or category.

2.4 Check Book Status

The student checks whether a book is available or already issued.

2.5 Pay Fine

If a book is returned late, the student pays the fine through the system.

2.6 Issue Request

The student sends a request to borrow a book from the library.

3. Faculty

3.1 Register

The faculty member registers in the system to use library services.

3.2 Login

The faculty member logs into the system.

3.3 Suggest Books

Faculty can recommend new books that should be added to the library.

3.4 Search Books

Faculty can search books in the library database.

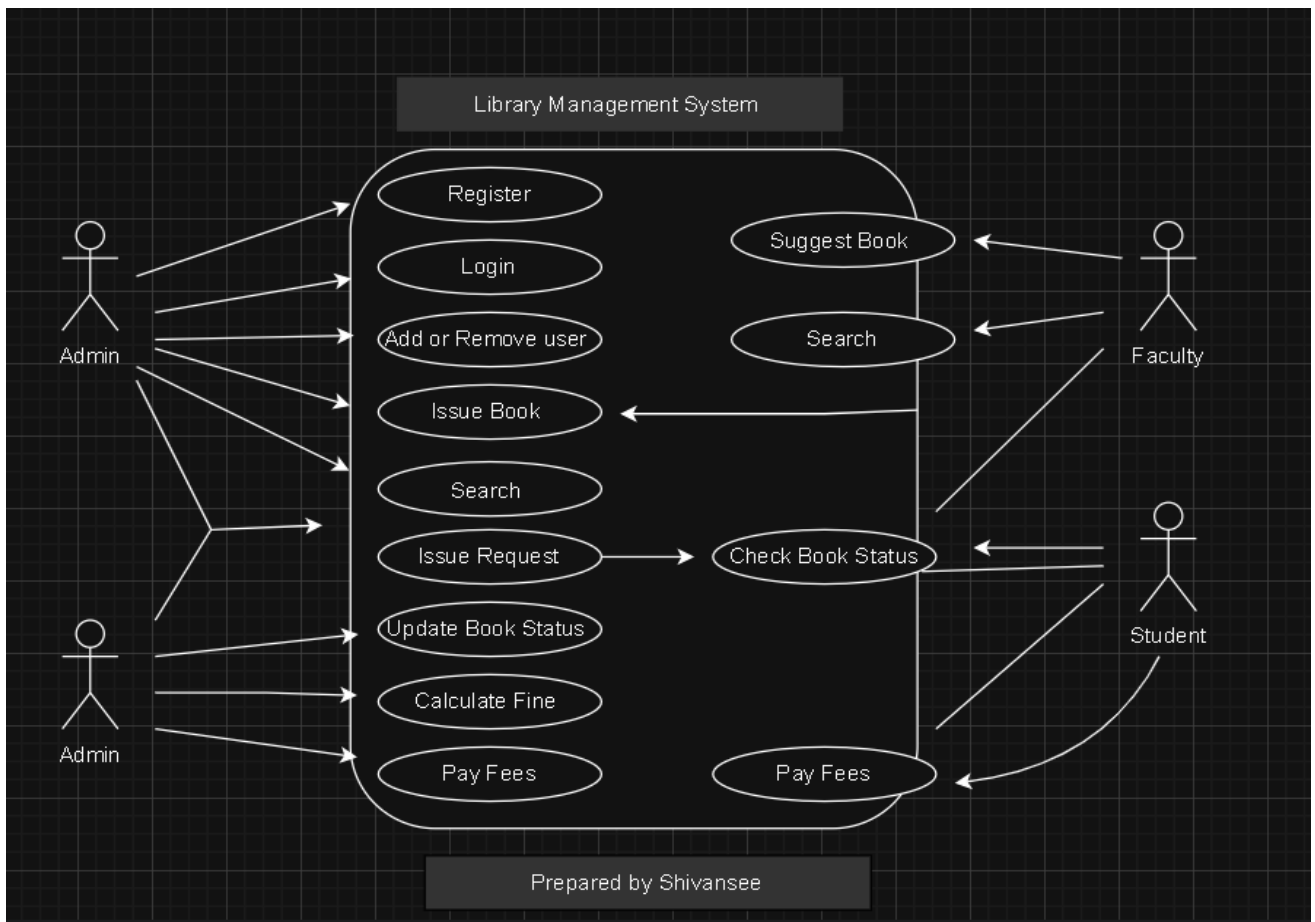
3.5 Issue Request

Faculty can request books for borrowing.

3.6 Check Book Status

Faculty can check whether the required book is available.

Figure 1: Use Case Diagram for Library Management System (LMS)



PRACTICAL – 2

Aim: To create Use Case Diagrams and define actors, preconditions, postconditions, and use case functions for Library Management System and Railway Ticket Booking System.

Introduction: Use Case Modeling is a technique used in software engineering to describe how users interact with a system. It shows the relationship between actors (users) and system functions.

Use case diagrams help in understanding system functionality and how different users use the system.

In this experiment, use case diagrams are created for:

1. Library Management System
2. Railway Ticket Booking System

Case Tool: StarUML

Scenario A: Library Management System

Actors:

1. Administrator
2. Student
3. Faculty

Preconditions:

These conditions must be satisfied before using the system.

1. The user must already be registered in the system.
2. The user must have valid login credentials.
3. The library database must contain book information.
4. The system must be connected to the database.

Postconditions:

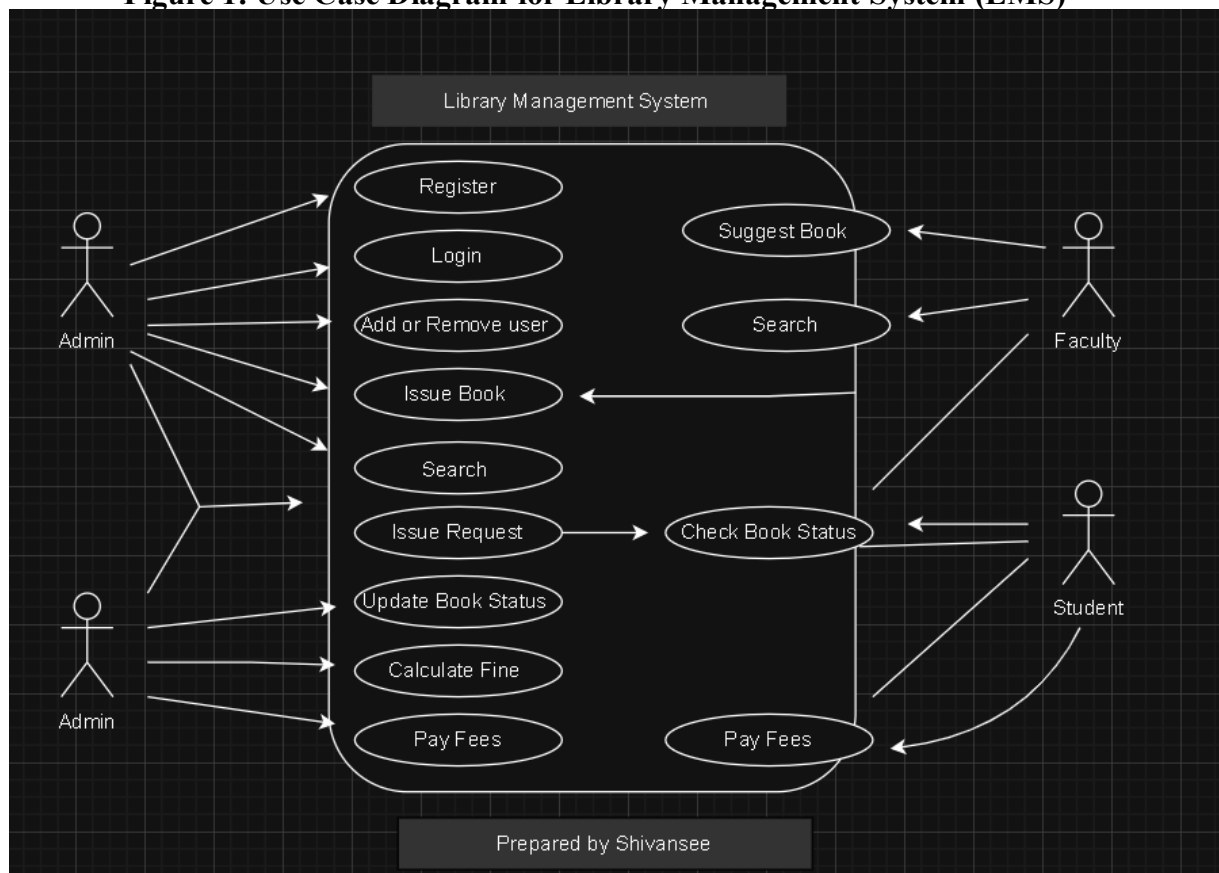
These are the results after completing an operation.

1. Book status is updated after issuing or returning.
2. User information is stored in the database.
3. Fine is calculated if a book is returned late.
4. All transactions are saved in the system database.

Use Case Functions:

1. **Register:** Allows a new user to create an account in the system.
2. **Login:** Allows users to log into the system using valid credentials.
3. **Search Books:** Users can search books using title, author, or category.
4. **Issue Book:** Admin issues books to users.
5. **Return Book:** Users return books and the system updates the record.
6. **Check Book Status:** Users check whether the book is available or issued.
7. **Calculate Fine:** Admin calculates the fine for late book returns.
8. **Pay Fine:** Users pay the fine for overdue books.

Figure 1: Use Case Diagram for Library Management System (LMS)



Scenario B: Railway Ticket Booking System

Actors:

1. Passenger
2. Administrator
3. Payment System

Preconditions:

1. Passenger must be registered in the system.
2. Passenger must log into the system before booking tickets.
3. Train schedule information must exist in the database.

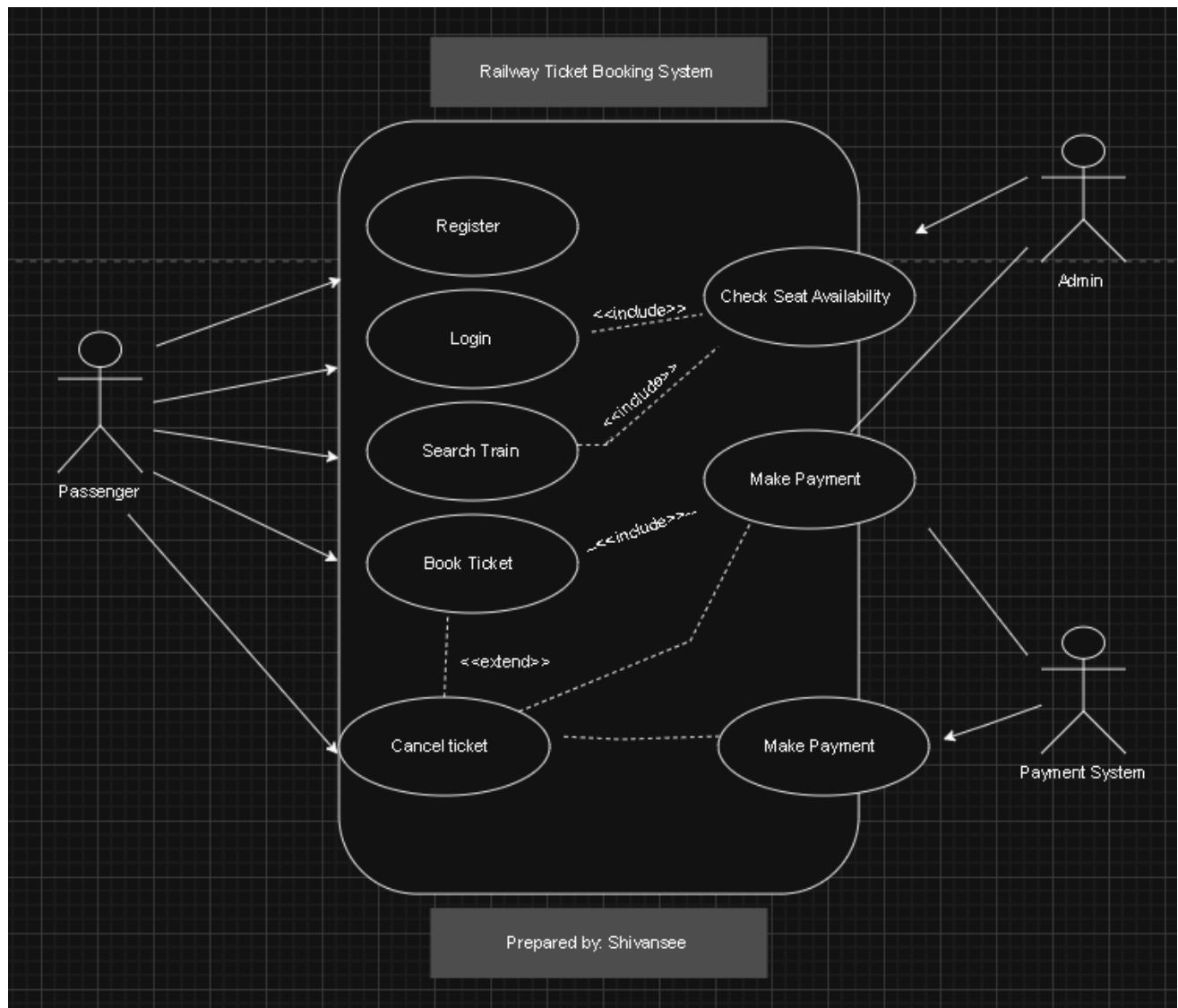
Postconditions:

1. Ticket is successfully booked after payment.
2. Seat availability is updated in the system.
3. Passenger receives booking confirmation.

Use Case Functions:

1. **Register:** Passenger creates an account in the system.
2. **Login:** Passenger logs into the system.
3. **Search Trains:** Passenger searches trains based on source, destination, and travel date.
4. **Check Seat Availability:** Passenger checks available seats in the train.
5. **Book Ticket:** Passenger books a ticket for the selected train.
6. **Make Payment:** Passenger completes the payment to confirm the booking.
7. **Cancel Ticket:** Passenger cancels a previously booked ticket.

Figure 2: Use Case Diagram for Railway Ticket Booking System



PRACTICAL – 3

Aim: To create Activity Diagrams to represent system workflow and dynamic behavior.

Introduction: An Activity Diagram is a UML diagram used to represent the workflow of a system. It shows the sequence of steps required to complete a process. Activity diagrams help in understanding how tasks are performed step by step in a system.

Case Tool: Draw.io

a. Activity Diagram: Issuing a Book in Library Management System

Actors:

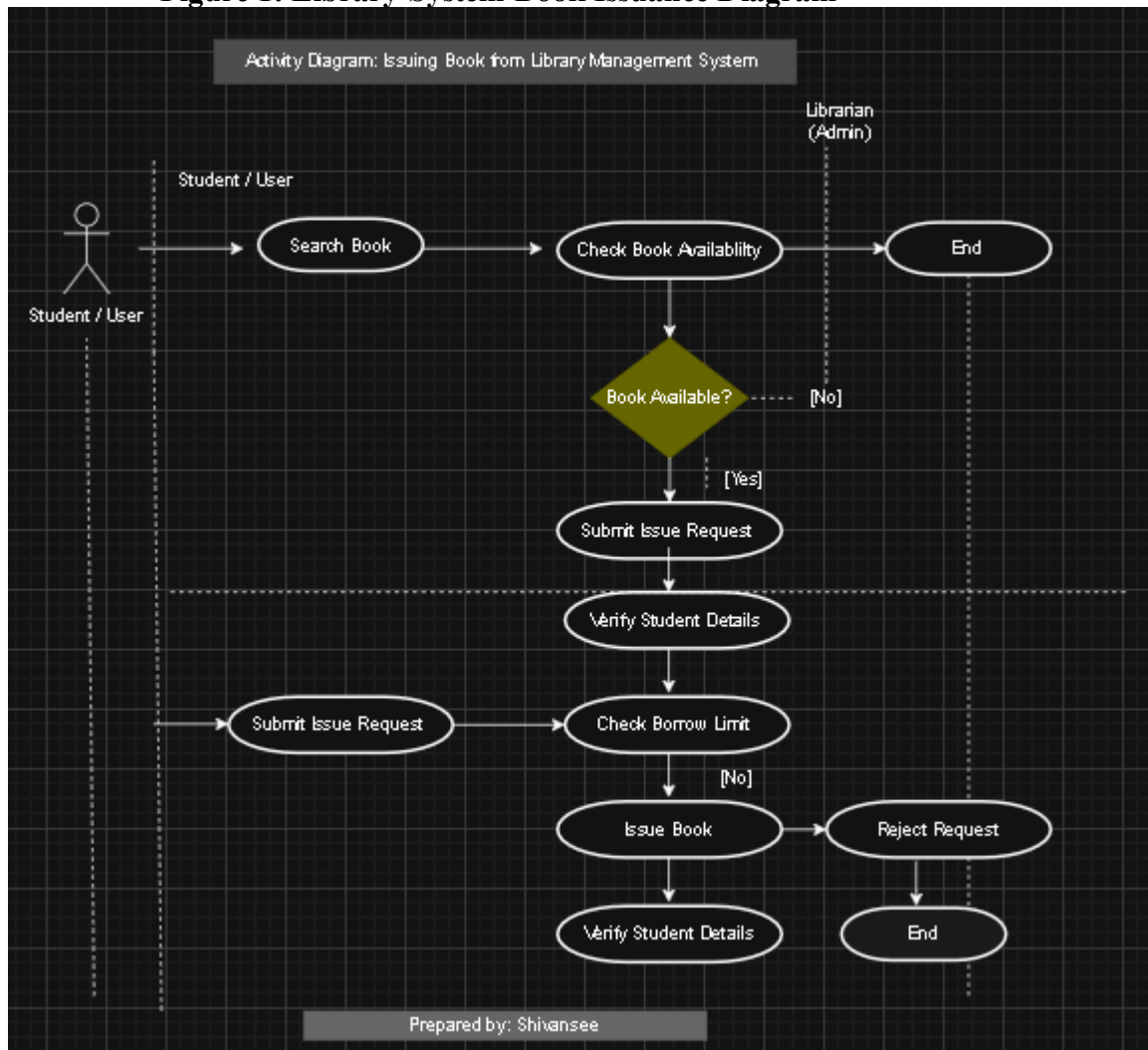
1. Student / User
2. Librarian (Admin)

Workflow Steps

1. Student searches for a book.
2. System checks whether the book is available.
3. If the book is not available, the process ends.
4. If the book is available, the student sends an issue request.
5. Librarian verifies student information.

6. System checks whether the borrowing limit is exceeded.
7. If the limit is exceeded, the request is rejected.
8. If the limit is valid, the librarian issues the book.
9. System updates the book status in the database.
10. System records the issue date and due date.
11. Process ends.

Figure 1: Library System Book Issuance Diagram



b. Activity Diagram: Opening Bank Account

Actors

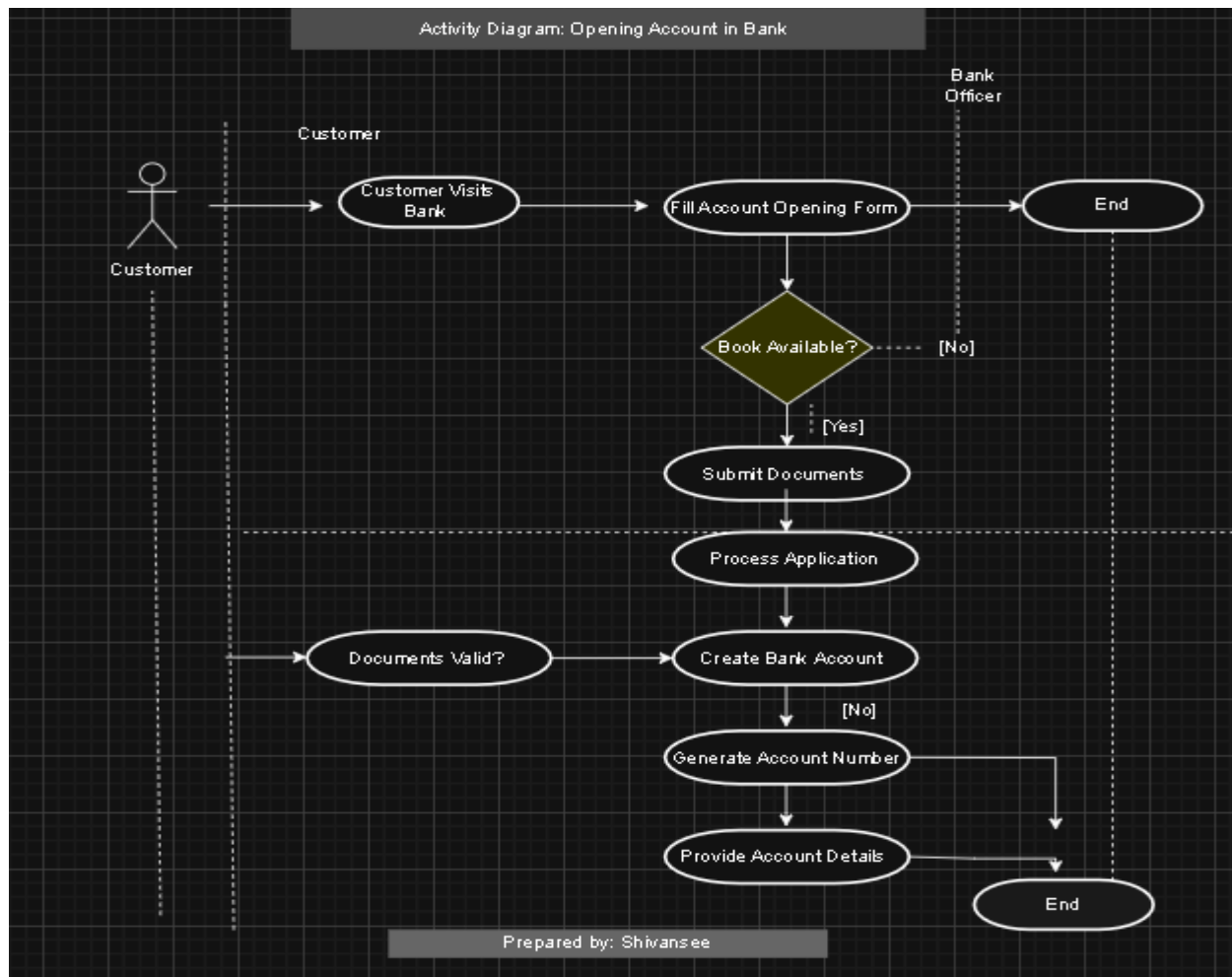
1. Customer
2. Bank Officer

Workflow Steps

1. Customer visits the bank or online portal.
2. Customer fills the account opening form.
3. Customer submits required documents such as ID proof and address proof.
4. Bank officer verifies the documents.
5. If documents are invalid, the request is rejected.
6. If documents are valid, the bank processes the request.

7. Bank creates a new account.
8. Account number is generated.
9. Customer receives account details.
10. Process ends.

Figure 2: Opening a Bank Account Diagram



PRACTICAL – 4

Aim: To design an Entity Relationship (ER) Diagram to represent the entities and relationships in Library Management System and Flight Ticket Booking System.

Introduction: An Entity Relationship (ER) Diagram is a graphical representation of entities, attributes, and relationships within a database system. It is used in database design to model how data is stored, connected, and managed within a system.

Case Tool: Draw.io

a. ER Diagram: Library Management System I.

Entities:

1. Student
2. Book
3. Librarian
4. Issue Record
5. Fine

II. Attributes:

1. Student

- Student_ID (Primary Key)
- Name
- Department
- Email
- Phone

2. Book

- Book_ID (Primary Key)
- Title
- Author
- Publisher
- Availability Status

3. Librarian

- Librarian_ID (Primary Key)
- Name
- Contact

4. Issue Record

- Issue_ID (Primary Key)
- Issue_Date
- Return_Date

5. Fine

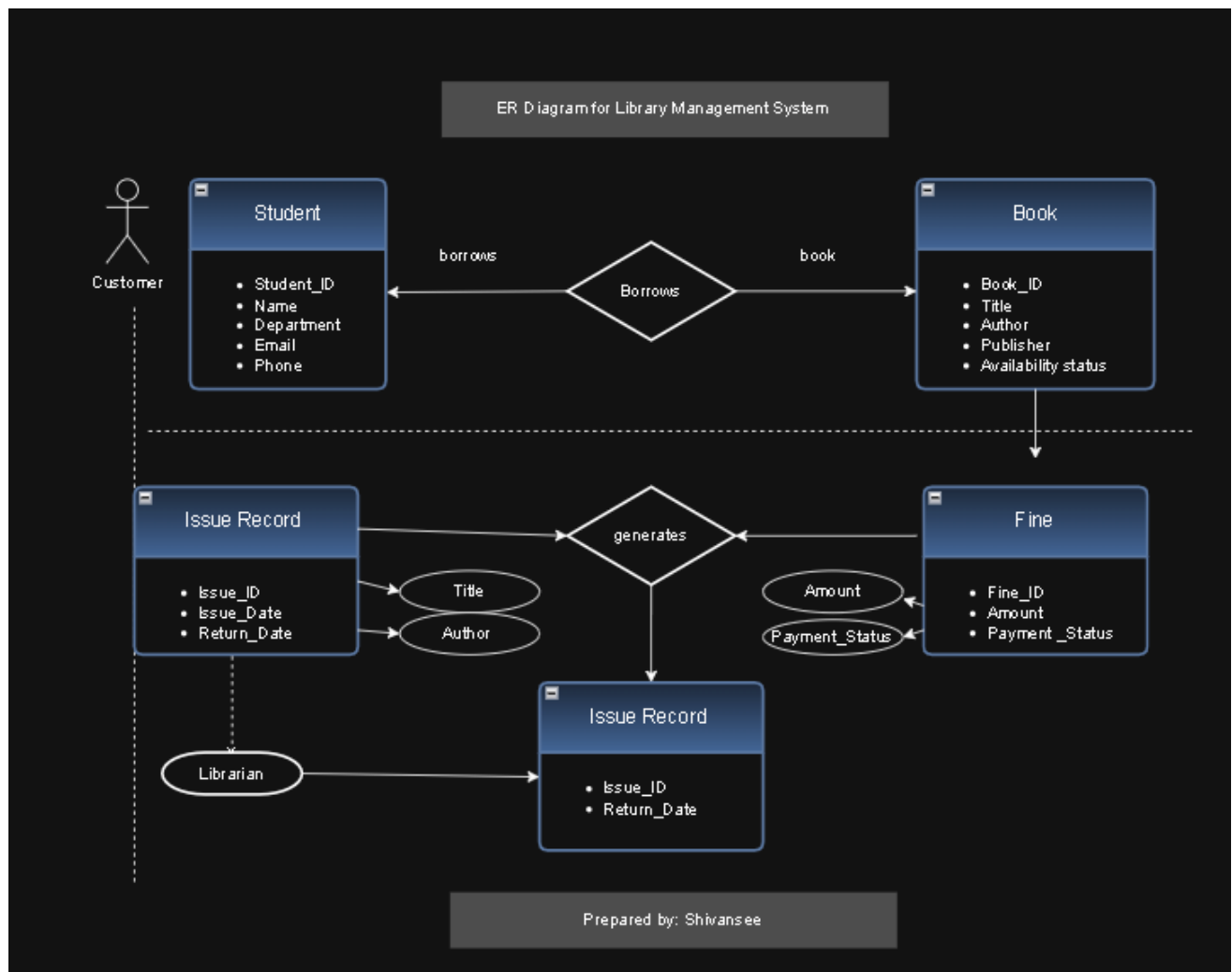
- Fine_ID (Primary Key)
- Amount
- Payment_Status

Relationship Identified:

1. Student borrows Book
2. Librarian manages Book

3. Book has Issue Record
4. Issue Record generates Fine

Figure 1: ER Diagram for Library Management System



b. ER Diagram: Flight Ticket Booking System

I. Entities

1. Passenger
2. Flight
3. Ticket
4. Payment
5. Airport

II. Attributes

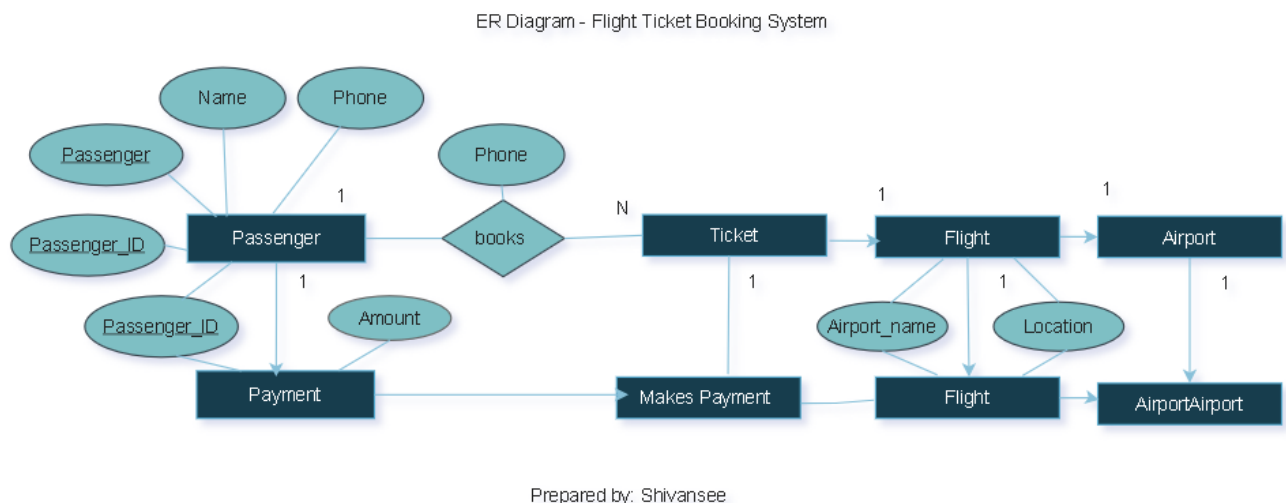
1. **Passenger**
 - Passenger_ID (Primary Key)
 - Name
 - Email
 - Phone
2. **Flight**
 - Flight_ID (Primary Key)

- Flight_Name
- Departure_Time
- Arrival_Time
- 3. **Ticket**
 - Ticket_ID (Primary Key)
 - Seat_Number
 - Booking_Date
- 4. **Payment**
 - Payment_ID (Primary Key)
 - Amount
 - Payment_Method
 - Payment_Status
- 5. **Airport**
 - Airport_ID (Primary Key)
 - Airport_Name
 - Location

Relationship Identified:

1. Passenger books Ticket
2. Ticket belongs to Flight
3. Passenger makes Payment
4. Flight departs from Airport
5. Flight arrives at Airport

Figure 2: ER Diagram for Flight Ticket Booking System



PRACTICAL – 5

Aim: To model object interactions over time using UML Sequence Diagrams for Validating Banking Transaction and E-commerce Checkout System.

Introduction: A Sequence Diagram is a type of UML diagram used to show how objects interact with each other in a particular sequence of time. It represents the flow of messages between different components in a system.

Case Tool: Draw.io

a. Sequence Diagram: Validating Banking Transaction

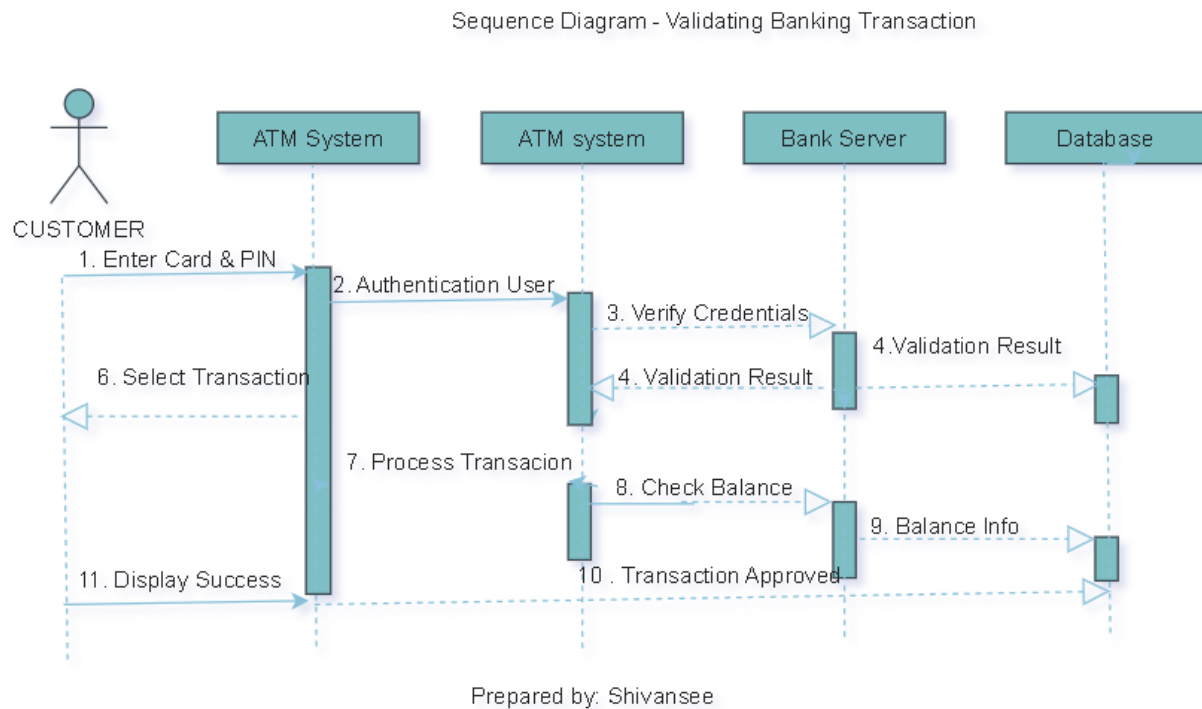
Actors Involved:

1. Customer
2. ATM / Banking Interface
3. Bank Server
4. Database

Workflow Steps:

1. Customer enters card and PIN.
2. ATM sends authentication request to Bank Server.
3. Bank Server checks credentials in Database.
4. Database returns verification result.
5. If credentials are valid → transaction options are displayed.
6. Customer selects transaction (withdraw / transfer).
7. ATM sends transaction request to Bank Server.
8. Bank Server verifies balance in Database.
9. Database sends balance status.
10. Bank Server approves transaction.
11. ATM displays success message.
12. Transaction completed.

Figure 1: Sequence Diagram for Validating Banking Transaction



b. Sequence Diagram: E-commerce Checkout **Actors**

Involved:

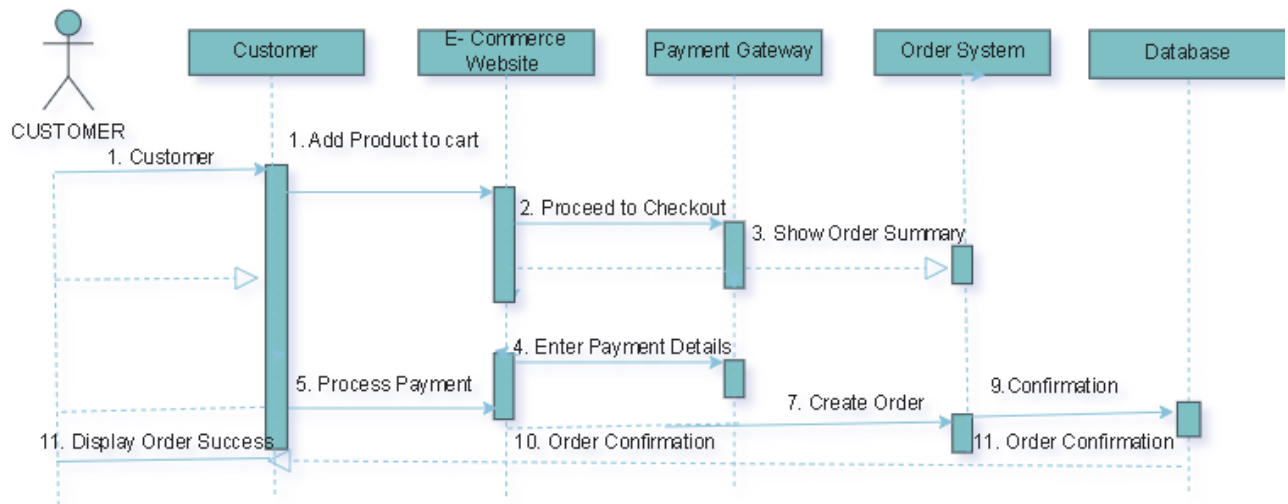
1. Customer
2. Website / Application
3. Payment Gateway
4. Order System
5. Database

Workflow Steps:

1. Customer adds products to cart.
2. Customer proceeds to checkout.
3. Website displays order summary.
4. Customer enters shipping and payment details.
5. Website sends payment request to Payment Gateway.
6. Payment Gateway verifies payment.
7. Payment Gateway sends payment status.
8. Order System creates order.
9. Database stores order details.
10. Website displays order confirmation.

Figure 2: Sequence Diagram for E-commerce Checkout System

Sequence Diagram- Validating Banking Transaction



Prepared by: Shivansee